Yanning DAI

PhD Student, King Abdullah University of Science and Technology (KAUST)

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Education

Ph.D. in Computer Science *Jun 2024 – Present*
King Abdullah University of Science and Technology (KAUST), Saudi Arabia GPA: 3.9/4.0
Advisor: Prof. Jürgen Schmidhuber

M.S. in Electrical Engineering *Sep 2020 - Jul 2023*
Beihang University, China GPA: 3.8/4.0
Advisor: Prof. Shuo Gao

B.E. in Electrical Engineering *Sep 2016 - Jul 2020*
Beihang University, China GPA: 3.8/4.0

Research Interests

Self-improving Agents, Reinforcement Learning, Robotic Learning

Selected Publications

1. Wearable-Informed Generative Digital Avatars Predict Task-Conditioned Post-Stroke Locomotion.
   Yanning Dai, Chenyu Tang, Ruizhi Zhang, Wenyu Yang, Yilan Zhang, Yuhui Wang, Junliang Chen, Xuhang Chen, Ruimou Xie, Yangyue Cao, Qiaoying Li, Jin Cao, Tao Li, Hubin Zhao, Yu Pan, Arokia Nathan, Xin Gao, Peter Smielewski, and Shuo Gao.
   Science Advances *(Accepted), 2026.*

2. Efficient Morphology–Control Co-Design via Stackelberg Proximal Policy Optimization.
   Yanning Dai*, Yuhui Wang*, Dylan R. Ashley, and Jürgen Schmidhuber.
   ICLR*, 2026.*

3. EvoAlloc: A Self-Evolving Resource Allocation Agent for Efficient Program Evolution.
   Yanning Dai, Yuhui Wang, Nanbo Li, Wenyi Wang, and Jürgen Schmidhuber.
   arXiv:2610.12086, 2026.

4. OASIS: Online Anchor-calibrated Self-Improvement System.
   Wenyi Wang, Firas Laakom, Li Nanbo, **Yanning Dai**, Yuhui Wang, Yifan Xu, Dmitrii Khizbullin, Piotr Piękos, Yilan Zhang, Yimeng Chen, Alfonso Amayuelas, Mingchen Zhuge, and Jürgen Schmidhuber.
   EMNLP Findings*, 2026.*

5. A Unified Framework for Rethinking Policy Divergence Measures in GRPO.
   Qingyuan Wu, Yuhui Wang, Simon Sinong Zhan, **Yanning Dai**, Shilong Deng, Sarra Habchi, Qi Zhu, Matthias Gallé, and Chao Huang.
   arXiv:2602.05494, 2026.

6. Predictive Semantic Safety: From Visual Physical Reasoning to Safety-Critical Control.
   Taekyung Kim*, Salem Fradi*, **Yanning Dai**, Mateusz Ostaszewski, and Jürgen Schmidhuber.
   arXiv:2609.34356, 2026.

7. Towards an Extremely Robust Baby Robot With Rich Interaction Ability for Advanced Machine Learning Algorithms.
   Mohannad Alhakami*, Dylan R. Ashley*, Joel Dunham*, **Yanning Dai**, Francesco Faccio, Eric Feron, and Jürgen Schmidhuber.
   IROS*, 2025.*

8. Tactile and Vision Perception for Intelligent Humanoids.
   Shuo Gao, **Yanning Dai**, and Arokia Nathan.
   ***Advanced Intelligent Systems**, 2022.* **[Cover Article]**.

9. Advanced Electronics and Artificial Intelligence: Must-have Technologies toward Human Body Digital Twins.
   Yanning Dai, Jiaqi Wang, and Shuo Gao.
   ***Advanced Intelligent Systems**, 2022.*

10. Wearable Systems based Gait Monitoring and Analysis.
    Shuo Gao, Junliang Chen, **Yanning Dai**, and Boyi Hu.
    ***Springer Nature**, 2022.* **[Book]**.

11. A Flexible Multi-Functional Smart Skin for Force, Touch Position, Proximity, and Humidity Sensing for Humanoid Robots.
    Yanning Dai and Shuo Gao.
    ***IEEE Sensors Journal**, 2021.*

12. A PVDF/Au/PEN Multifunctional Flexible Human–Machine Interface for Multidimensional Sensing and Energy Harvesting for the Internet of Things.
    Yanning Dai, Junliang Chen, Wenbin Tian, Lijun Xu, and Shuo Gao.
    ***IEEE Sensors Journal**, 2020.*

13. Multi-Functional Smart Skin for Multi-Dimensional Perception for Humanoid Robots.
    Yanning Dai and Shuo Gao.
    ***IEEE FLEPS**, 2020.*

*Equal contribution.

Experience

Research Intern *Jan 2024– May 2024*
Beijing Tsinghua Changgung Hospital, Tsinghua University

Built data-driven world models for patient-specific recovery dynamics on large-scale clinical datasets.

Research Assistant *Jun 2023– Jan 2024*
Advanced Sensing and Intelligent Systems Lab, Beihang University

Developed reinforcement learning and generative models for personalized digital human systems and behavior generation in stroke rehabilitation.

Research Intern *Jun 2022– Jun 2023*
National Engineering Research Center for Ophthalmic Diagnosis

Modeled patient-specific temporal dynamics of myopia progression from longitudinal ophthalmic data.

Selected Honors

1. Rising Stars in AI, KAUST AI Initiative (30 Awardees Worldwide) *Apr 2024*

2. National Scholarship, Ministry of Education of China (Top 5%) *Dec 2022*

3. The 3rd Class Award of Science and Technology, China Instrument and Meter Society (National S&T Award, Rank: 2/6) *Oct 2022*

4. China Instrument and Control Society Scholarship, 1st Class Award, China Instrument and Meter Society (Top Society Award, Top 1%) *Nov 2019*

Skills

Programming: Python, C++, C, C#, HTML, Shell, Matlab

Frameworks and Tools: PyTorch, TensorFlow, Transformers, Distributed Training (FSDP), Accelerated Inference (SGLang, vLLM), ROS, Robot Simulation (MuJoCo, Gazebo, Unity), Unitree SDK

Languages: English (Fluent), Mandarin (Native)

Teaching

School of Instrumentation and Optoelectronic Engineering, Beihang University:

Teaching Assistant, IntelliSense and Autonomous Systems *Spring 2023*
Teaching Assistant, Introduction to Smart Sensing Systems *Fall 2022*

Services

Conference Reviewer: NeurIPS, ICLR, EMNLP, ICRA, IEEE FLEPS.

Journal Reviewer: Expert Systems with Applications; IEEE Transactions on Neural Systems and Rehabilitation Engineering; IEEE Sensors Journal; IEEE Sensors Letters; Frontiers in Electronics.